Unit 4 · Session 2 · Thu, Nov 12, 2026

Endocrine Pharmacotherapy — Session 2

5 parts · every fact cited to its slide or textbook page. About 140 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

70 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

15 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

30 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. **Combined contraception** suppresses GnRH/FSH/LH; no estrogen with prior VTE/stroke, **smoker ≥ 35**, **migraine with aura**, estrogen-sensitive cancer, uncontrolled hypertension → progestin-only. Inducers (rifampin) need backup. S5; S6

2. **Tamoxifen** blocks breast but activates endometrium (report bleeding); **raloxifene** activates bone, blocks endometrium; both raise VTE. **Finasteride** lowers DHT and halves PSA. S7; S8; S9

3. **Insulin**: rapid (lispro, aspart), short (regular — the only IV insulin), NPH, basal (glargine, detemir, degludec); main risk **hypoglycemia** (rule of 15; beta-blockers mask it). S12; S13

4. **Metformin** first-line (avoid eGFR < 30, hold for contrast, B12); **sulfonylureas** highest oral hypoglycemia; **TZDs** fluid/heart failure; **SGLT2** genital infections, euglycemic DKA, cardiorenal benefit. S14; S15; S16

5. **GLP-1 RAs** (weight loss; no MTC/MEN2) and **DPP-4 inhibitors** (weight neutral) work glucose-dependently — low hypoglycemia; older adult who falls → DPP-4 or SGLT2. S18; S19; S21

6. **Hyperparathyroidism** → cinacalcet (watch hypocalcemia); **hypoparathyroidism** → calcitriol + calcium (watch hypercalcemia). S25; S27

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO4.1 Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders

IO4.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs

IO4.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 5 · Putting It All Together: Endocrine Toolkit

IO4.4 Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.

Part 1 · Gonadal Hormones & Inhibitors
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders
Part 5 · Putting It All Together: Endocrine Toolkit

IO4.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.

Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders

IO4.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 5 · Putting It All Together: Endocrine Toolkit

SO1 Describe the mechanisms and clinical uses of gonadal hormone agents, including combined hormonal contraception, SERMs, and androgens/antiandrogens.

Part 1 · Gonadal Hormones & Inhibitors

SO2 Classify glucose-lowering drugs by mechanism and site of action, and compare agents on efficacy, hypoglycemia risk, and key adverse effects.

Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs

SO3 Distinguish hyperparathyroidism from hypoparathyroidism and identify the drug strategy used for each.

Part 4 · Parathyroid Disorders

SO4 Identify key contraindications, adverse effects, and drug interactions throughout.

Part 1 · Gonadal Hormones & Inhibitors
Part 3 · Non-Insulin Glucose-Lowering Drugs

SO5 Apply the full Endocrine pharmacology toolkit to integrated clinical scenarios, and complete a cumulative recall quiz spanning both sessions.

Part 5 · Putting It All Together: Endocrine Toolkit

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 4 · SESSION 2 OF 2** — Endocrine Pharmacotherapy. **Session 2: Reproductive Hormones, Diabetes, and Bone Mineral Homeostasis | Thursday, November 12 | 8:30–11:00 AM** S1
- Recap: Session 1 (Nov 10) covered the hypothalamic-pituitary axis, thyroid/antithyroid drugs, and adrenocorticosteroid pharmacology. Session 1 gave us the axis. S1 (notes); S2; S2 (notes)
- Today the same negative-feedback logic explains contraception, then the many ways to lower glucose, finishing with the two opposite parathyroid problems — then integration and the cumulative recall quiz. S1 (notes); S2 (notes)
- Today picks up with gonadal hormones (Section 4) and runs through diabetes, parathyroid disorders, integration, and the cumulative recall quiz. S2 (notes)
- **Appendix 1 abbreviation key: PO = by mouth; SC = subcutaneous; IM = intramuscular; IV = intravenous; ER / XL = extended-release; BID = twice daily; mcg = microgram; U-100 etc. = units of insulin per mL; IUD = intrauterine device.** S36; S36 (notes)
- **DCIS = ductal carcinoma in situ; VTE / DVT / PE = venous thromboembolism / deep vein thrombosis / pulmonary embolism; MI = myocardial infarction; BPH = benign prostatic hyperplasia; PSA = prostate-specific antigen; PCOS = polycystic ovary syndrome.** S36; S36 (notes)
- **DKA = diabetic ketoacidosis; eGFR / CrCl = estimated glomerular filtration rate / creatinine clearance; CKD = chronic kidney disease; NPH = neutral protamine Hagedorn; SGLT2 = sodium-glucose cotransporter 2; DPP-4 = dipeptidyl peptidase-4; GLP-1 = glucagon-like peptide 1.** S36; S36 (notes)
- **MEN2 = multiple endocrine neoplasia type 2; CaSR = calcium-sensing receptor; PTH = parathyroid hormone; NYHA = New York Heart Association; CV = cardiovascular; CYP = cytochrome P450; QT = QT interval on ECG.** S36; S36 (notes)
- **APPENDIX 1** gives **Formulation, dose, indications, adverse effects, contraindications and toxicity** — your study reference: know the dose logic, the serious adverse effects, the contraindication that changes your decision, and the toxicity you would need to recognize. S36; S36 (notes); S37-66
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions and rescue. **Listed for context; not profiled in detail.** S67; S67 (notes)
- **Original figures in this deck:** Figures 11, 15, 19, 21 and 22 (Combined Hormonal Contraception: Suppressing the HPG Axis; Pancreatic β-Cell Insulin Secretion: Secretagogue & Incretin Targets; Glucose-Lowering Drugs by Site of Action; S68
- HbA1c-Lowering Efficacy vs. Hypoglycemia Risk by Class; Parathyroid Disorders: Two Opposite Problems, Two Drug Strategies) were generated by the course director using the AI tool Claude

— original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. S68; S68 (notes)

- **Mechanism illustrations** (AI-generated, illustrative, not reproduced from copyrighted sources; schematic and not to scale): combined pills (Figure 12, slide 6); SERMs (Figure 13, slide 8); testosterone and 5-alpha-reductase inhibitors (Figure 14, slide 9); insulin signaling (Figure 16, slide 15); S68
- metformin (Figure 17, slide 17); SGLT2 inhibitors (Figure 18, slide 18); incretin drugs (Figure 20, slide 21); cinacalcet and the calcium-sensing receptor (Figure 23, slide 32). **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S68

SLIDE DISCREPANCIES

- **Rapid-acting insulin onset:** slides say about 15 minutes; the textbook gives 5–15 minutes. Same practical rule: dose with the meal. S12; p. 791

1 PART 1 · DRUG CLASS

Gonadal Hormones & Inhibitors

SO1 SO4 IO4.1 IO4.2 IO4.4 IO4.6

Class

- **Suppressing, replacing, or selectively modulating the reproductive axis.** Today the same negative-feedback logic explains contraception. S3; S3 (notes)
- **Figure 11. Original illustrative diagram (not from Katzung): exogenous estrogen and progestin suppress GnRH pulsatility and the FSH/LH surge, preventing the dominant-follicle development and LH surge required for ovulation.** S4; S4 (notes)
- This is the same negative-feedback template from Figure 1 (Session 1), now applied to the HPG axis. S4 (notes)

Prototype + other drugs named

- Combined oral contraceptives (ethinyl estradiol plus a progestin); progestin-only: norethindrone "minipill," depot medroxyprogesterone, levonorgestrel IUD; SERMs tamoxifen, raloxifene; testosterone; finasteride; spironolactone. S5; S7

Mechanism of action

- **Mechanism: estrogen + progestin suppress GnRH/FSH/LH, preventing ovulation (Figure 11); progestin also thickens cervical mucus and thins the endometrium.** S5; S6 (notes)
- Chain (Figure 12): ethinyl estradiol plus progestin enter the bloodstream → hypothalamus lowers GnRH pulses; pituitary lowers FSH and LH → no follicle maturation and no LH surge, so no ovulation → result: pregnancy prevented (over 99% with perfect use). S6; S4 (notes)
- SERMs: tamoxifen or raloxifene bind the estrogen receptor → the receptor shape recruits different co-factors in each tissue. S8
- Breast: blocked (antagonist), so cancer growth is slowed. Bone: activated (agonist), so bone density is kept.

Indications

- Combined contraception is available as oral pills, transdermal patch, or vaginal ring. S5
- Tamoxifen — antagonist in breast tissue (breast cancer treatment/prevention), agonist in endometrium (raises endometrial cancer risk). Tissue-selective activity is the defining feature of this class. S7; S7 (notes)
- Raloxifene — antagonist in breast/endometrium, agonist in bone (osteoporosis prevention) — ties back to Unit 2 bone pharmacology; prevents vertebral fracture; not used to treat breast cancer. S7; S8
- Testosterone — replacement for hypogonadism; monitor hematocrit (erythrocytosis risk) and lipids. Finasteride — 5-α-reductase inhibitor; reduces conversion of testosterone to DHT (BPH, androgenetic alopecia). S7

Endometrium: tamoxifen activates (cancer risk); raloxifene blocks. S8
- Testes make testosterone → 5-alpha-reductase converts it to DHT in prostate and skin → DHT drives prostate growth and hair loss → finasteride blocks 5-alpha-reductase → lower DHT: smaller prostate, less hair loss. S9; S9 (notes)

- Spironolactone — antiandrogen effect at high doses; used off-label for hirsutism/acne (same drug as the mineralocorticoid antagonist from Session 1): same molecule, different receptor, different clinical use. S7; S7 (notes); S9

Adverse effects (boxed warnings first)

- Combined pills: risks include venous thromboembolism, hypertension and stroke. Common effects: nausea, breast tenderness, breakthrough bleeding. S6
- Tamoxifen raises endometrial cancer and VTE risk and causes hot flashes; patients should report vaginal bleeding. Raloxifene does not increase endometrial cancer, but raises VTE risk and hot flashes. S8; S8 (notes)
- Finasteride lowers PSA by about half, so double the PSA value when screening; sexual side effects occur. S9; S9 (notes)

Contraindications

- **Contraindications: history of VTE/stroke, estrogen-sensitive cancer, uncontrolled hypertension, migraine with aura, and smokers age 35 or older (increased thromboembolic risk).** Liver disease too. S5; S5 (notes)
- Smoking at age 35+ substantially raises thromboembolic/cardiovascular risk with estrogen — the basis of the contraindication. S5; SG U4-R10
- "Smoker over 35" and "migraine with aura" are the classic exam triggers for choosing a progestin-only method — smokers 35 or older, migraine with aura and prior VTE rule out estrogen. S5 (notes); S6
- Progestin-only options (e.g., the "minipill," depot medroxyprogesterone (bone density warning), levonorgestrel IUD) avoid estrogen-related thromboembolic risk and are preferred when estrogen is contraindicated. S5; S6; S6 (notes)
- Both SERMs are contraindicated with prior VTE. Finasteride is teratogenic: pregnant women should not handle crushed or broken tablets. Testosterone: avoid in prostate cancer. S8; S9; S9 (notes)

Drug interactions

- Enzyme inducers (rifampin, carbamazepine, phenytoin, St. John's wort, some antiepileptics) lower contraceptive efficacy; use backup contraception. S5 (notes); S6; S67

Pharmacokinetics

Not covered in materials

- CYP2D6 inhibitors (paroxetine, fluoxetine) blunt tamoxifen activation. S8 (notes); S67

Monitoring

- Testosterone replacement: monitor hematocrit, PSA and lipids. S9; S9 (notes)

CLINICAL PEARL / WHY

- The contraindication list is the highest-yield content in contraception. S5 (notes)

Step by step

Can this patient take estrogen? S5; S6

Prior VTE or stroke? → Smoker aged 35 or older? → Migraine with aura? →

Estrogen-sensitive cancer, uncontrolled HTN, liver disease? →

Any yes → progestin-only (minipill, DMPA, levonorgestrel IUD)

Figure 11 · Combined hormonal contraception suppresses the HPG axis. Original illustrative diagram (not from Katzung): exogenous estrogen and progestin feed back on the hypothalamus and pituitary, suppressing GnRH pulses and the FSH/LH surge, so no dominant follicle and no ovulation. S4; S4 (notes)

Figure 12 · Combined pills: ovulation block and backup barriers. Schematic (illustrative) — Figure 12. Teal: hormone entry; blue: brain and ovary signaling; grey-green: local cervical and endometrial effects; red: outcome. Arrows show negative feedback: no ovulation, thick mucus, thin endometrium. S6; S6 (notes)

Figure 13 · SERMs: one receptor, different tissue effects. Schematic (illustrative) — Figure 13. Teal: receptor binding; blue and grey-green: tissue responses; green check = agonist, red X = antagonist; red outcome = risk in endometrium. Tamoxifen activates the endometrium (cancer risk); raloxifene blocks it. S8; S8 (notes)

Figure 14 · Testosterone, DHT and 5-alpha-reductase inhibitors. Schematic (illustrative) — Figure 14. Teal: start; blue: enzyme and hormone steps; green-grey: blocking drug; red: outcome, lower DHT. Finasteride blocks 5-alpha-reductase: smaller prostate, less hair loss, PSA roughly halved. S9; S9 (notes)

Contraceptive profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Combined oral contraceptives (ethinyl estradiol plus a progestin)	Estrogen plus progestin	Tablets (21/28-day packs); patch; vaginal ring	Ethinyl estradiol 20-35 mcg plus a progestin PO daily; patch weekly for 3 weeks; ring 3 weeks in, 1 week out	Contraception, cycle control, dysmenorrhea, acne
Norethindrone (progestin-only pill)	Progestin	Tablet 0.35 mg	0.35 mg PO daily at the same time, with no pill-free days	Contraception when estrogen is contraindicated, breastfeeding
Medroxyprogesterone depot (Depo-Provera)	Injectable progestin	IM 150 mg/mL; SC 104 mg/0.65 mL	150 mg IM (or 104 mg SC) every 12-13 weeks	Contraception, endometriosis pain, endometrial protection
Levonorgestrel IUD (Mirena)	Intrauterine progestin	Intrauterine system 52 mg (about 20 mcg/day)	One system placed in the uterus; effective up to 8 years for contraception	Contraception, heavy menstrual bleeding, endometrial protection

How to tell them apart. Sources: S37; S38; S39; S40

Contraceptive profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Combined oral contraceptives (ethinyl estradiol plus a progestin)	Nausea, breast tenderness, breakthrough bleeding, headache	VTE or stroke history, smoker 35 or older, migraine with aura, estrogen-sensitive cancer, uncontrolled hypertension, liver disease	VTE, stroke and MI risk; hyperkalemia with drospirenone; rifampin and some antiepileptics lower efficacy
Norethindrone (progestin-only pill)	Irregular bleeding, headache, acne, breast tenderness	Breast cancer, undiagnosed vaginal bleeding, liver tumor	Efficacy falls if a dose is more than 3 hours late; ectopic pregnancy risk if it fails
Medroxyprogesterone depot (Depo-Provera)	Irregular bleeding, weight gain, delayed return of fertility	Breast cancer, pregnancy, unexplained vaginal bleeding	Boxed warning: loss of bone mineral density; limit prolonged use; VTE
Levonorgestrel IUD (Mirena)	Irregular spotting, amenorrhea, ovarian cysts, pelvic pain	Pregnancy, pelvic infection, uterine anomaly, breast cancer	Uterine perforation, expulsion, infection; check strings

How to tell them apart. Sources: S37; S38; S39; S40

SERM and androgen profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Tamoxifen (Nolvadex)	Selective estrogen-receptor modulator	Tablet 10/20 mg	20 mg PO daily for 5-10 years in breast cancer	Breast cancer treatment and risk reduction, DCIS
Raloxifene (Evista)	Selective estrogen-receptor modulator	Tablet 60 mg	60 mg PO daily	Postmenopausal osteoporosis treatment and prevention, breast cancer risk reduction
Testosterone (cypionate, gel, patch)	Androgen	IM cypionate or enanthate 100-200 mg/mL; gel; patch; pellets	Cypionate 100-200 mg IM every 2 weeks (or 50-100 mg weekly); gel 50-100 mg daily	Male hypogonadism, delayed puberty
Finasteride (Proscar, Propecia)	5-alpha-reductase inhibitor	Tablet 5 mg (BPH); 1 mg (hair loss)	5 mg PO daily for BPH; 1 mg PO daily for male-pattern hair loss	BPH, androgenetic alopecia

How to tell them apart. Sources: S41; S42; S43; S44

SERM and androgen profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Tamoxifen (Nolvadex)	**Hot flashes, vaginal discharge, menstrual irregularity**	**History of DVT or PE, pregnancy, warfarin (for risk-reduction use)**	**Endometrial cancer, VTE, stroke, cataracts; CYP2D6 inhibitors (paroxetine, fluoxetine) reduce activation**
Raloxifene (Evista)	**Hot flashes, leg cramps, peripheral edema**	**VTE history, pregnancy**	**Boxed warning: VTE and fatal stroke risk; does not stimulate the endometrium**
Testosterone (cypionate, gel, patch)	**Acne, erythrocytosis, sleep apnea, gynecomastia, testicular atrophy**	**Prostate or breast cancer, polycythemia, pregnancy**	**Erythrocytosis, thrombosis, secondary exposure with gels; monitor hematocrit, PSA, lipids**
Finasteride (Proscar, Propecia)	**Decreased libido, erectile dysfunction, reduced ejaculate, gynecomastia**	**Pregnancy and women who may be pregnant (do not handle crushed tablets), hypersensitivity**	**Lowers PSA by about half; reports of depression; teratogenic to a male fetus**

How to tell them apart. Sources: S41; S42; S43; S44

WHY IT HAPPENS

- Why does a smoker over 35 get a progestin-only method? Estrogen adds clotting risk on top of smoking and age — the progestin-only options carry no estrogen. S5; S6

HIGH-YIELD

- No estrogen with prior VTE/stroke, smoker ≥ 35, migraine with aura, estrogen-sensitive cancer, uncontrolled hypertension → progestin-only. S5; S6
- Tamoxifen: endometrial agonist (report bleeding); raloxifene: bone agonist, endometrium blocked; both VTE. S7; S8

2 PART 2 · DRUG CLASS

Insulin

SO2 IO4.1 IO4.2 IO4.5 IO4.6

Class

- **Many mechanisms, one shared goal: glycemic control without excess hypoglycemia.** Today we cover the many ways to lower glucose. S10; S10 (notes)
- **Figure 15. Original illustrative diagram (not from Katzung): glucose entry raises ATP, closing K-ATP channels and triggering depolarization and calcium-dependent insulin release;** S11; S11 (notes)
- **sulfonylureas/meglitinides force channel closure directly, while GLP-1 receptor agonists and DPP-4 inhibitors amplify glucose-dependent release via the incretin pathway.** S11
- This is the mechanistic foundation for the insulin secretagogue and incretin-based drug classes that follow — the "glucose-dependent" nature of incretin-based therapy is exactly why these agents carry lower hypoglycemia risk than sulfonylureas. S11 (notes)
- TB Diabetes mellitus has four categories: type 1 (beta-cell destruction, absolute insulin deficiency), type 2, other specific types, and gestational diabetes (about 7% of US pregnancies). p. 789; p. 790
- TB Diagnosis: fasting plasma glucose ≥126 mg/dL or HbA1c ≥6.5%, confirmed by repeat testing. p. 790

Prototype + other drugs named

- Insulin lispro, aspart (rapid); regular (short); NPH (intermediate); glargine, detemir, degludec (long-acting). S12

Mechanism of action

- **Mechanism: exogenous insulin activates the insulin receptor, promoting cellular glucose uptake and suppressing hepatic gluconeogenesis.** S12

Indications

- Rapid-acting (lispro, aspart) — mealtime dosing; onset ~15 minutes. Short-acting (regular) — onset ~30 minutes. S12

- Chain (Figure 16): insulin binds the insulin receptor on muscle, fat and liver cells → receptor autophosphorylation starts the signal cascade → GLUT4 moves to the cell surface so glucose enters. S13; S13 (notes)
- Then the liver stores glycogen and stops making glucose → blood glucose falls (hypoglycemia is the main risk). S13

- Intermediate-acting (NPH) — cloudy suspension; more variable absorption. Long-acting (glargine, detemir, degludec) — basal coverage; relatively peakless profile. S12
- Rapid-acting is given with meals; regular 30 minutes before. NPH peaks at about 4–10 hours; glargine, detemir and degludec are basal. Only regular insulin is given IV. S12 (notes); S13
- Sort any named insulin into rapid, short, intermediate or long-acting before discussing regimens — the scaffold for every insulin question. S12 (notes)
- **TB** Glucagon (IM, SC, IV, or 3-mg nasal powder) treats severe hypoglycemia when oral glucose is impossible; place the patient on their side (vomiting). p. 789

Adverse effects (boxed warnings first)

- **Primary adverse effect across all insulins: hypoglycemia — counsel patients on recognition and treatment (rapid-acting carbohydrate).** S12
- Adverse effects: hypoglycemia, weight gain, lipohypertrophy, hypokalemia. Never share pens. S13
- **TB** Rotate injection sites to avoid lipohypertrophy; insulin allergy (IgE) is rare. p. 794; p. 795

Contraindications

Not covered in materials

Drug interactions

- Beta-blockers can mask hypoglycemia warning symptoms. Glucagon and dextrose: rescue for severe hypoglycemia. S12 (notes); S13; S67

Pharmacokinetics

- **TB** Insulin preparations: lispro/aspart onset 5–15 min, 3–4 h duration; regular 30–60 min, 6–8 h; NPH 2–4 h, peak 6–7 h; glargine about 24 h, peakless. p. 791; p. 793
- **TB** Glargine is acidic (pH 4.0) — never mix it in the same syringe with other insulins. p. 793; p. 794

Monitoring

- Rule of 15 for hypoglycemia: 15 g carbohydrate, recheck in 15 minutes. S12 (notes); S13

CLINICAL PEARL / WHY

Not covered in materials

Figure 15 · β-cell insulin secretion: secretagogue and incretin targets. Original illustrative diagram (not from Katzung): glucose → ATP → K-ATP channels close → depolarization → Ca^{2+} → insulin release. Sulfonylureas/meglitinides force closure directly (any glucose); GLP-1 agonists and DPP-4 inhibitors amplify release only when glucose is high. S11; S11 (notes)

Figure 16 · Insulin: receptor signaling and time-action profiles. Schematic (illustrative) — Figure 16. Teal: insulin binding; blue: intracellular signaling and glucose uptake; grey-green: liver; red: outcome. Right panel shows onset, peak and duration by insulin type: rapid (lispro, aspart), short (regular), NPH (peak ~4–10 h), basal (glargine, detemir, degludec). S13; S13 (notes)

Insulin by time-action (S12)

Type	Examples	Onset / profile	Use
Rapid-acting	Lispro, aspart	~15 min	Mealtime
Short-acting	Regular	~30 min	30 min before meals; only IV insulin
Intermediate	NPH	Cloudy; peak ~4–10 h; variable	Basal/peak coverage
Long-acting	Glargine, detemir, degludec	Relatively peakless	Basal coverage

How to tell them apart. Sources: S12; S13

Insulin profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Insulin lispro (Humalog)	Rapid-acting insulin analog	SC or IV vial or pen 100 or 200 units/mL	Mealtime dose 0-15 minutes before eating; total daily insulin about 0.5-1 unit/kg; onset about 15 minutes	Type 1 and type 2 diabetes
Insulin aspart (NovoLog)	Rapid-acting insulin analog	SC or IV vial and pen cartridges 100 units/mL	Mealtime dose 0-15 minutes before eating; part of total daily insulin 0.5-1 unit/kg	Type 1 and type 2 diabetes, pump therapy
Regular insulin (Humulin R, Novolin R)	Short-acting human insulin	Vial 100 units/mL; U-500 concentrate; IV	SC 30 minutes before meals; IV infusion for DKA about 0.1 units/kg/hour	Diabetes mellitus, DKA and hyperkalemia (IV)
NPH insulin (Humulin N, Novolin N)	Intermediate-acting human insulin	Vial or pen 100 units/mL (cloudy suspension)	SC once or twice daily; basal start about 0.1-0.2 units/kg; peaks at 4-10 hours	Type 1 and type 2 diabetes, steroid-induced hyperglycemia
Insulin glargine (Lantus, Basaglar, Toujeo)	Long-acting basal insulin analog	SC pen or vial 100 or 300 units/mL	Start 10 units SC daily (or 0.1-0.2 units/kg) and titrate to fasting glucose	Type 1 and type 2 diabetes (basal)
Insulin detemir (Levemir)	Long-acting basal insulin analog	SC pen or vial 100 units/mL	10 units (or 0.1-0.2 units/kg) once or twice daily, titrated to fasting glucose	Type 1 and type 2 diabetes (basal)
Insulin degludec (Tresiba)	Ultra-long-acting basal insulin analog	SC pen 100 or 200 units/mL	10 units (or 0.1-0.2 units/kg) once daily; dosing interval flexible if at least 8 hours apart	Type 1 and type 2 diabetes (basal)

How to tell them apart. Sources: S45; S46; S47; S48; S49; S50; S51

Insulin profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Insulin lispro (Humalog)	Hypoglycemia, weight gain, lipohypertrophy, hypokalemia	Hypoglycemia, hypersensitivity	Severe hypoglycemia, hypokalemia; dose errors with concentrated pens; rotate sites
Insulin aspart (NovoLog)	Hypoglycemia, weight gain, injection-site reactions	Hypoglycemia, hypersensitivity	Severe hypoglycemia; hypokalemia; do not mix in a pump with other insulins

Drug	Adverse effects	Contraindications	Toxicity / overdose
Regular insulin (Humulin R, Novolin R)	**Hypoglycemia, weight gain, hypokalemia (IV)**	**Hypoglycemia**	**Only regular insulin may be given IV; U-500 dosing errors; monitor potassium**
NPH insulin (Humulin N, Novolin N)	**Hypoglycemia (nocturnal), weight gain, variable absorption**	**Hypoglycemia**	**Nocturnal hypoglycemia at the peak; mix suspension gently before use**
Insulin glargine (Lantus, Basaglar, Toujeo)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia, hypersensitivity**	**Hypoglycemia; do not mix with other insulins; U-300 is not interchangeable unit for unit**
Insulin detemir (Levemir)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia**	**Hypoglycemia; duration depends on dose; availability varies, so verify**
Insulin degludec (Tresiba)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia**	**Lower nocturnal hypoglycemia than glargine; steady state in 3-4 days; check pen strength**

How to tell them apart. Sources: S45; S46; S47; S48; S49; S50; S51

MNEMONIC (MEMORY AID)

- Rule of 15: 15 g fast carbohydrate, recheck in 15 minutes. S13

HIGH-YIELD

- Insulin's main risk is hypoglycemia; beta-blockers mask its warning signs; only regular insulin goes IV. S12; S13

3 PART 3 · DRUG CLASS

Non-Insulin Glucose-Lowering Drugs

SO2 SO4 IO4.1 IO4.2 IO4.3 IO4.4 IO4.5

Class

- Four non-incretin oral cards: metformin, sulfonylureas/meglitinides, TZDs, SGLT2 inhibitors; then the incretin classes (GLP-1 RAs, DPP-4 inhibitors) and the site-of-action map. S14; S14 (notes)
- **Figure 19. Original illustrative diagram (not from Katzung): glucose-lowering drug classes act at distinct sites — pancreatic β-cell, liver, muscle/fat, kidney, gut, and the incretin axis — each converging on overall blood glucose control.** S17; S17 (notes)

Prototype + other drugs named

- Metformin; glipizide, glyburide (sulfonylureas); repaglinide (meglitinide); pioglitazone (TZD); empagliflozin (SGLT2); exenatide, liraglutide, semaglutide (GLP-1 RA); sitagliptin, linagliptin (DPP-4). S14; S18

Mechanism of action

- Metformin decreases hepatic gluconeogenesis: enters hepatocytes through OCT1 → inhibits mitochondrial complex I and activates AMPK → gluconeogenesis falls → insulin sensitivity improves in muscle → fasting glucose falls without hypoglycemia. S14; S15; S15 (notes)
- Sulfonylureas / meglitinides force K-ATP channel closure — insulin secretagogues (glipizide, glyburide, repaglinide). TZDs (pioglitazone) increase peripheral insulin sensitivity. S14
- Match each drug class to its mechanism/site: metformin — ↓ hepatic gluconeogenesis; sulfonylureas — force K-ATP channel closure (insulin secretagogue); TZDs — ↑ peripheral insulin sensitivity; SGLT2 inhibitors — block renal glucose reabsorption. SG U4-M; SG U4-R11

Indications

- Metformin: first-line for type 2 diabetes; no weight gain and low hypoglycemia risk. S14; S15; S15 (notes)
- SGLT2 inhibitors: cardiorenal benefit — benefits in heart failure and chronic kidney disease, beyond glucose lowering. S14; S16; S16 (notes)
- **Figure 21. Original illustrative comparison (illustrative relative values, not from Katzung): insulin and sulfonylureas show the greatest HbA1c-lowering efficacy but also carry the highest hypoglycemia risk,** S20
- **while metformin, DPP-4 inhibitors, and SGLT2 inhibitors show a more favorable efficacy-to-hypoglycemia-risk balance.** S20
- Figure 21 is the quantified summary of the drug-class discussion — it answers "why is metformin still first-line": not because it is

- GLP-1 receptor agonists — glucose-dependent insulin release, slowed gastric emptying; DPP-4 inhibitors — prevent endogenous GLP-1 breakdown (incretin pathway). Sulfonylureas carry the highest hypoglycemia risk (glucose-independent insulin release). SG U4-M; SG U4-R11
- SGLT2 inhibitors block renal glucose reabsorption: glucose is filtered at the glomerulus → SGLT2 in the proximal tubule normally reabsorbs about 90% → empagliflozin blocks SGLT2. S14; S16
- Glucose and water are lost in the urine (glucosuria) → glucose, weight and blood pressure fall; infection risk rises. S16; S16 (notes)
- GLP-1 receptor agonists: glucose-dependent insulin release, slow gastric emptying, promote satiety/weight loss. DPP-4 inhibitors prevent breakdown of endogenous GLP-1, prolonging its glucose-dependent effect. S18; S18 (notes)
- Incretin chain (Figure 20): food in the gut releases GLP-1 → GLP-1 acts on the beta cell: glucose-dependent insulin release, less glucagon → DPP-4 normally breaks GLP-1 down quickly. S19; S19 (notes)
- Sitagliptin blocks DPP-4; semaglutide and liraglutide mimic GLP-1 → lower glucose, slower gastric emptying and less appetite. S19

the most potent, but because of its favorable risk profile. S20 (notes)

- **TB** Semaglutide lowers HbA1c 1.5–1.8%; liraglutide 3 mg and semaglutide 2.4 mg are approved for weight loss without diabetes. p. 800
- **TB** SGLT2 inhibitors lower HbA1c 0.5–1%, cause 2–5 kg weight loss, and reduce heart-failure events (dapagliflozin hazard ratio 0.74). p. 801; p. 802

Adverse effects (boxed warnings first)

- Sulfonylureas: highest hypoglycemia risk among oral agents. TZDs: fluid retention/heart failure, weight gain. S14
- Metformin: GI upset is common — start low and take with food; vitamin B12 deficiency with long-term use; monitor B12 levels. S15; S15 (notes)
- SGLT2 inhibitors: genital yeast (mycotic) infections, volume depletion, euglycemic DKA — hold when ill or before surgery; rare Fournier gangrene. S14; S16; S16 (notes)
- GLP-1 RAs: nausea/GI upset, vomiting, pancreatitis risk and gallstones. DPP-4

Contraindications

- Metformin: avoid in significant renal impairment — lactic acidosis risk; avoid when eGFR is below 30; hold for contrast dye with reduced kidney function (iodinated contrast). S14; S15; S15 (notes); S67
- GLP-1 RAs: contraindicated with personal or family history of medullary thyroid carcinoma or MEN2 (thyroid C-cell warning). S18; S18 (notes); S19
- SGLT2 inhibitors: avoid for glycemic effect when eGFR is low. S16
- **TB** Metformin: full dose at eGFR 45–60, cautious at 30–45, contraindicated below

inhibitors are generally weight-neutral, well tolerated, with low hypoglycemia risk. S18; S18 (notes); S19

- **TB** Sulfonylureas: hypoglycemia and weight gain; elderly patients are at particular risk. p. 796; p. 797
- **TB** Acarbose (an alpha-glucosidase inhibitor, a drug that affects absorption of glucose): treat hypoglycemia with glucose, not sucrose. p. 799

30; avoid in decompensated liver failure; check renal function at least annually. p. 798

- **TB** Thiazolidinediones are contraindicated in NYHA class III and IV heart failure. p. 799

Drug interactions

- Do not combine a GLP-1 RA with a DPP-4 inhibitor. Gemfibrozil raises repaglinide levels (CYP2C8). S19; S67

Pharmacokinetics

- **TB** Sitagliptin: reduce dose in renal impairment (50 mg at eGFR 30–50, 25 mg below 30). Exenatide is not approved below eGFR 30. p. 800; p. 801

Monitoring

Not covered in materials

CLINICAL PEARL / WHY

- Case: a 72-year-old with type 2 diabetes on metformin monotherapy, HbA1c above goal, lives alone, prior fall. **Which class of add-on therapy best balances efficacy and safety for this patient, and why?** S21; S21 (notes)
- **Given the fall history and the risks of hypoglycemia-induced falls, a DPP-4 inhibitor or SGLT2 inhibitor (low intrinsic hypoglycemia risk) is generally preferable to a sulfonylurea in an older adult living alone —** S21
- **matching drug choice to patient-specific risk, not just HbA1c-lowering potency.** S21
- This vignette applies Figure 21 directly to a geriatric safety scenario: "most potent" and "most appropriate" are not always the same answer. S21 (notes)
- Drug names reveal the class: **-gliptin** DPP-4 inhibitor (sitagliptin, linagliptin; low hypoglycemia risk); **-glutide** GLP-1 receptor agonist (liraglutide, semaglutide; weight loss, GI upset). S28; S28 (notes)
- **-gliflozin** SGLT2 inhibitor (empagliflozin; genital infections, euglycemic DKA); **-glitazone** thiazolidinedione (pioglitazone; fluid retention, heart failure); **-glinide** meglitinide (repaglinide; mealtime insulin secretagogue). S28
- **gli- (prefix)** sulfonylurea (glipizide, glyburide; K-ATP closure, hypoglycemia); **-ifene** SERM (tamoxifen, raloxifene; tissue-selective estrogen action); **-steride** 5-alpha-reductase inhibitor (finasteride; lowers DHT); **-gestrel** progestin (levonorgestrel). S28
- **-calcet** calcimimetic (cinacalcet; activates the CaSR, lowers PTH). A stem is a memory aid, not a rule: where no shared stem exists, learn the class name instead. S28

- **TB** SGLT2 inhibitors: not recommended for glycemic control below eGFR 45, contraindicated below 30 (for glucose). p. 802

Step by step

Add-on after metformin S16; S18; S21

HbA1c above goal on metformin → Older, lives alone, falls? → avoid sulfonylurea → Heart failure or CKD → SGLT2 inhibitor →

Weight loss wanted → GLP-1 RA (no MTC/MEN2) →

Low-risk, weight neutral → DPP-4 inhibitor

Figure 17 · Metformin: hepatic glucose output and AMPK. Schematic (illustrative) — Figure 17. Teal: drug entry; blue: AMPK signaling and glucose output; grey-green: muscle; short arrow = reduced gluconeogenesis; red: outcome — fasting glucose falls without hypoglycemia. S15; S15 (notes)

Figure 18 · SGLT2 inhibitors: glucose lost in the urine. Schematic (illustrative) — Figure 18. Teal: filtration; blue: tubule transport; green-grey: drug block; red: outcome, glucose lost in urine — lower glucose, weight and BP, with genital infection and euglycemic DKA risks. S16; S16 (notes)

Figure 19 · Glucose-lowering drugs by site of action. Original illustrative diagram (not from Katzung): β-cell (sulfonylureas, meglitinides), liver (metformin), muscle/fat (TZDs), kidney (SGLT2 inhibitors), gut and incretin axis (GLP-1 RAs, DPP-4 inhibitors) — all converging on blood glucose. S17; S17 (notes)

Figure 20 · Incretin drugs: gut, pancreas and brain. Schematic (illustrative) — Figure 20. Teal: gut signal; blue: beta-cell and enzyme steps; green-grey: drugs; red: clinical outcome. GLP-1 RAs mimic GLP-1; DPP-4 inhibitors protect it — lower glucose, slower gastric emptying, less appetite. S19; S19 (notes)

Figure 21 · HbA1c lowering vs. hypoglycemia risk. Original illustrative comparison (illustrative relative values, not from Katzung): insulin and sulfonylureas lower HbA1c most but carry the highest hypoglycemia risk; metformin, DPP-4 inhibitors and SGLT2 inhibitors balance efficacy and hypoglycemia risk more favorably. S20; S20 (notes)

Glucose-lowering classes (S14; S18)

Class	Example	Mechanism	Key risk
Metformin	Metformin	↓ Hepatic gluconeogenesis	GI, B12; lactic acidosis — avoid eGFR < 30
Sulfonylureas / meglitinides	Glipizide, glyburide, repaglinide	Force K-ATP closure	Highest oral hypoglycemia risk
TZDs	Pioglitazone	↑ Peripheral insulin sensitivity	Fluid retention/heart failure, weight gain
SGLT2 inhibitors	Empagliflozin	Block renal glucose reabsorption	Genital mycotic infection, euglycemic DKA
GLP-1 RAs	Exenatide, liraglutide, semaglutide	Glucose-dependent insulin; slow emptying	GI, pancreatitis; MTC/MEN2
DPP-4 inhibitors	Sitagliptin, linagliptin	Protect endogenous GLP-1	Weight-neutral; low hypoglycemia

How to tell them apart. Sources: S14; S18

Oral glucose-lowering profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Metformin (Glucophage)	**Biguanide**	**Tablet 500/850/1,000 mg; ER tablet; oral solution**	**Start 500 mg PO daily or BID with meals, titrate to 1,000 mg BID (max 2,550 mg/day)**	**Type 2 diabetes, prediabetes, PCOS (off-label)**
Glipizide (Glucotrol)	**Sulfonylurea (second generation)**	**Tablet 5/10 mg; XL 2.5/5/10 mg**	**5 mg PO daily 30 minutes before breakfast, titrate (max 40 mg/day; XL 20 mg)**	**Type 2 diabetes**
Glyburide (Diabeta, Micronase)	**Sulfonylurea (second generation)**	**Tablet 1.25/2.5/5 mg; micronized 1.5/3/6 mg**	**2.5-5 mg PO daily with breakfast, titrate (max 20 mg/day)**	**Type 2 diabetes**
Repaglinide (Prandin)	**Meglitinide**	**Tablet 0.5/1/2 mg**	**0.5-2 mg PO before each meal (skip if the meal is skipped), max 16 mg/day**	**Type 2 diabetes (mealtime)**
Pioglitazone (Actos)	**Thiazolidinedione (PPAR-gamma agonist)**	**Tablet 15/30/45 mg**	**15-30 mg PO daily, max 45 mg**	**Type 2 diabetes**

Drug	Class	Formulation	Adult dose	Indications
Empagliflozin (Jardiance)	SGLT2 inhibitor	Tablet 10/25 mg	10 mg PO once daily in the morning, may increase to 25 mg	Type 2 diabetes with cardiorenal benefit, heart failure, CKD

How to tell them apart. Sources: S52; S53; S54; S55; S56; S57

Oral glucose-lowering profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Metformin (Glucophage)	Nausea, diarrhea, metallic taste, vitamin B12 deficiency	eGFR below 30, acute metabolic acidosis, hypersensitivity	Lactic acidosis (renal impairment, contrast, alcohol, sepsis); hold around iodinated contrast; monitor B12
Glipizide (Glucotrol)	Hypoglycemia, weight gain, nausea, rash	Type 1 diabetes, DKA, severe liver disease; caution in sulfonamide allergy	Prolonged hypoglycemia (older adults, renal impairment); alcohol and beta-blockers worsen or mask it
Glyburide (Diabeta, Micronase)	Hypoglycemia, weight gain, nausea	Type 1 diabetes, DKA, bosentan co-use	Active metabolites cause prolonged hypoglycemia; avoid in older adults and CKD
Repaglinide (Prandin)	Hypoglycemia, weight gain, upper respiratory infection	Type 1 diabetes, DKA, gemfibrozil co-use	Hypoglycemia; gemfibrozil markedly raises levels (CYP2C8); CYP3A4 interactions
Pioglitazone (Actos)	Edema, weight gain, fractures, upper respiratory infection	NYHA class III-IV heart failure (boxed), active bladder cancer, liver disease	Heart failure exacerbation, fractures in women, possible bladder cancer; monitor weight and edema
Empagliflozin (Jardiance)	Genital yeast infections, UTI, volume depletion	Dialysis, hypersensitivity; not indicated in type 1 diabetes	Euglycemic DKA (hold when ill or fasting), Fournier gangrene, hypotension

How to tell them apart. Sources: S52; S53; S54; S55; S56; S57

Incretin drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Exenatide (Byetta, Bydureon)	GLP-1 receptor agonist	SC pen 5/10 mcg (Byetta); 2 mg weekly (Bydureon)	Byetta 5 mcg BID within 60 minutes before meals, then 10 mcg BID; Bydureon 2 mg SC once weekly	Type 2 diabetes

Drug	Class	Formulation	Adult dose	Indications
Liraglutide (Victoza, Saxenda)	GLP-1 receptor agonist	SC pen 18 mg/3 mL	Victoza 0.6 mg daily for 1 week, then 1.2-1.8 mg daily; Saxenda up to 3 mg daily	Type 2 diabetes, CV risk reduction, chronic weight management (Saxenda)
Semaglutide (Ozempic, Wegovy, Rybelsus)	GLP-1 receptor agonist	SC pen weekly; oral tablet 3/7/14 mg	Ozempic 0.25 mg weekly for 4 weeks, then 0.5-2 mg weekly; Rybelsus 3 mg daily, then 7-14 mg	Type 2 diabetes, CV risk reduction, obesity (Wegovy)
Sitagliptin (Januvia)	DPP-4 inhibitor	Tablet 25/50/100 mg	100 mg PO daily; 50 mg if eGFR 30-45; 25 mg if below 30	Type 2 diabetes
Linagliptin (Tradjenta)	DPP-4 inhibitor	Tablet 5 mg	5 mg PO daily; no renal adjustment	Type 2 diabetes

How to tell them apart. Sources: S58; S59; S60; S61; S62

Incretin drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Exenatide (Byetta, Bydureon)	Nausea, vomiting, diarrhea, injection-site nodules	Medullary thyroid cancer or MEN2 (Bydureon), CrCl below 30, severe gastroparesis	Pancreatitis, thyroid C-cell tumor warning (Bydureon), renal injury
Liraglutide (Victoza, Saxenda)	Nausea, vomiting, diarrhea, constipation, headache	Personal or family medullary thyroid cancer, MEN2, pancreatitis history	Boxed thyroid C-cell tumors, pancreatitis, gallbladder disease, dehydration with kidney injury
Semaglutide (Ozempic, Wegovy, Rybelsus)	Nausea, vomiting, diarrhea, constipation, abdominal pain	Personal or family medullary thyroid cancer, MEN2, hypersensitivity	Boxed thyroid C-cell tumors, pancreatitis, retinopathy complications, gallbladder disease, aspiration risk with anesthesia
Sitagliptin (Januvia)	Upper respiratory infection, headache, nasopharyngitis	Hypersensitivity (anaphylaxis, angioedema)	Pancreatitis, severe joint pain; low hypoglycemia risk when used alone
Linagliptin (Tradjenta)	Nasopharyngitis, cough, diarrhea	Hypersensitivity	Pancreatitis, bullous pemphigoid, arthralgia; eliminated mainly through bile

How to tell them apart. Sources: S58; S59; S60; S61; S62

SLIDE DISCREPANCY

- **Metformin and eGFR**
- Slides: Avoid when eGFR is below 30. S15

- Textbook: Full use at 45–60, cautious at 30–45 (lower doses), contraindicated below 30. p. 798

WHY IT HAPPENS

- Why do DPP-4 inhibitors rarely cause hypoglycemia? The incretin effect only boosts insulin when glucose is already high; sulfonylureas push insulin out regardless. S11; S35

HIGH-YIELD

- Metformin first-line (no weight gain, low hypoglycemia); avoid eGFR < 30; hold for contrast; B12. S14; S15
- Older adult with falls → DPP-4 or SGLT2, not a sulfonylurea. S21

4 PART 4 · DRUG CLASS

Parathyroid Disorders

SO3 IO4.1 IO4.4 IO4.5

Class

- **A brief recap of bone pharmacology, then a new angle: PTH itself.** Unit 2 covered osteoporosis drug therapy in depth: bisphosphonates, denosumab, teriparatide, and raloxifene (a SERM, also discussed again this session). S22; S23
- That therapy targets bone remodeling directly, generally in patients with normal parathyroid function. **This section takes a different angle: what happens — and what drugs are used — when parathyroid hormone secretion itself is the primary problem.** S23; S23 (notes)
- **If you need a refresher on bisphosphonate dosing or denosumab safety, revisit the Unit 2, Session 2 materials — that content is not repeated here.** S23
- This section is intentionally NOT re-teaching osteoporosis pharmacotherapy — it is new content on parathyroid pathology, so don't expect bisphosphonate-level detail here. S23 (notes)
- **Figure 22. Original illustrative diagram (not from Katzung): hyperparathyroidism (excess PTH, high calcium) is treated with cinacalcet to lower PTH,** S24
- **while hypoparathyroidism (deficient PTH, low calcium) is treated with calcitriol and oral calcium, sometimes with recombinant PTH 1-84 in severe cases.** S24
- Figure 22 is the organizing concept for the section — two opposite pathologies, two opposite drug strategies, both targeting the same hormone axis. S24 (notes)

Prototype + other drugs named

- Cinacalcet (calcimimetic); calcitriol (active vitamin D) plus calcium; recombinant human PTH 1-84. S25

Mechanism of action

- Cinacalcet — a calcimimetic; activates the calcium-sensing receptor on parathyroid cells, reducing PTH secretion. S25
- Chain (Figure 23): parathyroid cells sense calcium through the CaSR → low calcium keeps PTH release high → cinacalcet makes the receptor more sensitive to calcium. S26
- The cell reads calcium as high and lowers PTH → PTH and serum calcium fall. S26; S26 (notes)
- Calcitriol (active vitamin D) plus oral calcium — mainstay for hypoparathyroidism, since PTH deficiency impairs vitamin D activation; calcitriol bypasses the PTH activation step. S25; S25 (notes)

Indications

- Cinacalcet: used for secondary hyperparathyroidism (e.g., in chronic kidney disease on dialysis) and parathyroid carcinoma. S25; S26
- Recombinant human PTH 1-84 — reserved for patients not adequately controlled on calcium/calcitriol alone. S25

Adverse effects (boxed warnings first)

- Cinacalcet adverse effects: nausea, hypocalcemia and QT prolongation, so monitor calcium. S26

Contraindications

Not covered in materials

Drug interactions

- Cinacalcet is a CYP2D6 inhibitor. Thiazides raise calcium, loops lower it. Vitamin D (cholecalciferol) replaces or supports calcitriol in deficiency. S26; S67

Pharmacokinetics

Not covered in materials

Monitoring

- Hyperparathyroidism — cinacalcet (lowers PTH secretion); hypoparathyroidism — calcitriol (bypasses the PTH-dependent vitamin D activation step) plus oral calcium. SG U4-R13
- Opposite monitoring targets: watch for hypocalcemia with cinacalcet, hypercalcemia/hypercalciuria with

calcitriol/PTH replacement — a symmetry worth drawing out. S25; S25 (notes); S26

CLINICAL PEARL / WHY

- Case: a patient with chronic kidney disease is found to have an elevated PTH level with low-normal serum calcium. A second, unrelated patient who underwent recent thyroidectomy develops perioral numbness and low serum calcium with a low PTH level. S27
- **How does drug therapy differ between these two patients?** S27
- **The CKD patient has secondary hyperparathyroidism and may benefit from cinacalcet to reduce PTH secretion.** S27
- **The post-thyroidectomy patient has surgical hypoparathyroidism (inadvertent parathyroid gland injury) and needs calcitriol plus calcium replacement — the opposite pharmacologic strategy for the opposite underlying problem.** S27; S27 (notes)
- Post-thyroidectomy hypoparathyroidism is a classic, testable scenario and a bridge back to thyroid surgery from Session 1 (antithyroid drugs can also precede surgical management). S27 (notes)

Step by step

Which parathyroid drug? S25; S27

Check PTH and calcium →

High PTH (CKD, dialysis) → cinacalcet; watch hypocalcemia →

Low PTH, low calcium (post-thyroidectomy) → calcitriol + calcium →

Refractory → recombinant PTH 1-84 → Watch hypercalcemia on replacement

Figure 22 · Parathyroid disorders: two opposite strategies. Original illustrative diagram (not from Katzung): hyperparathyroidism (excess PTH, high calcium) → cinacalcet lowers PTH; hypoparathyroidism (deficient PTH, low calcium) → calcitriol plus oral calcium, sometimes recombinant PTH 1-84. S24; S24 (notes)

Figure 23 · Cinacalcet: tricking the calcium-sensing receptor. Schematic (illustrative) — Figure 23. Teal: sensing; blue: PTH steps; green-grey: drug; red: outcome, PTH and calcium fall. Cinacalcet sensitizes the CaSR so the cell reads calcium as high. S26; S26 (notes)

Hyperparathyroidism vs. hypoparathyroidism therapy (S25)

	Hyperparathyroidism	Hypoparathyroidism
Drug	Cinacalcet (calcimimetic)	Calcitriol + oral calcium; PTH 1-84 if refractory
How	Activates CaSR → ↓ PTH	Bypasses the PTH-dependent vitamin D activation
Typical patient	CKD on dialysis; parathyroid carcinoma	Post-thyroidectomy
Monitor for	Hypocalcemia	Hypercalcemia / hypercalciuria

How to tell them apart. Sources: S25; S26; S27

Parathyroid and calcium profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Cinacalcet (Sensipar)	**Calcimimetic (CaSR activator)**	**Tablet 30/60/90 mg**	**30 mg PO daily with food, titrate every 2-4 weeks (up to 180 mg/day) by PTH and calcium**	**Secondary hyperparathyroidism on dialysis, parathyroid carcinoma, primary hyperparathyroidism when surgery is not possible**
Calcitriol (Rocaltrol)	**Active vitamin D (1,25-dihydroxy)**	**Capsule 0.25/0.5 mcg; solution 1 mcg/mL; IV**	**Hypoparathyroidism: 0.25-2 mcg PO daily, titrated to calcium**	**Hypoparathyroidism, CKD secondary hyperparathyroidism, hypocalcemia**
Recombinant PTH 1-84 (Natpara)	**Parathyroid hormone replacement**	**SC pen cartridges 25/50/75/100 mcg**	**50 mcg SC daily in the thigh, adjusted by 25 mcg up to 100 mcg**	**Hypoparathyroidism not controlled on calcium and vitamin D**
Calcium carbonate or citrate	**Calcium supplement**	**Carbonate 500-1,250 mg tablets; citrate tablets**	**1,000-2,000 mg elemental calcium daily in divided doses (500-600 mg per dose)**	**Hypocalcemia, hypoparathyroidism, osteoporosis**

How to tell them apart. Sources: S63; S64; S65; S66

Parathyroid and calcium profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Cinacalcet (Sensipar)	**Nausea, vomiting, diarrhea, paresthesias, hypocalcemia**	**Hypocalcemia at start**	**Severe hypocalcemia (QT prolongation, seizures); CYP2D6 inhibitor; monitor calcium and PTH**
Calcitriol (Rocaltrol)	**Hypercalcemia, hypercalciuria, nausea, constipation**	**Hypercalcemia, vitamin D toxicity**	**Hypercalcemia, kidney stones, calcification; monitor calcium, phosphate, urine calcium**

Drug	Adverse effects	Contraindications	Toxicity / overdose
Recombinant PTH 1-84 (Natpara)	**Hypercalcemia, hypocalcemia, paresthesias, headache**	**Hypersensitivity; caution with osteosarcoma risk factors**	**Boxed osteosarcoma warning; severe hypocalcemia if stopped abruptly; availability is restricted, so verify current status**
Calcium carbonate or citrate	**Constipation, bloating, hypercalcemia**	**Hypercalcemia, hypercalciuria, kidney stones**	**Hypercalcemia (especially with calcitriol); carbonate needs food and acid (citrate for PPI users); separate from levothyroxine**

How to tell them apart. Sources: S63; S64; S65; S66

WHY IT HAPPENS

- Why calcitriol, not plain vitamin D, after thyroidectomy? Activating vitamin D needs PTH; calcitriol is already active. S25; S35

HIGH-YIELD

- High PTH (CKD) → cinacalcet, watch hypocalcemia; low PTH (post-thyroidectomy) → calcitriol + calcium, watch hypercalcemia. S25; S27

5 PART 5 · TOPIC

Putting It All Together: Endocrine Toolkit

SO5 IO4.3 IO4.4 IO4.6

Key definitions

- **Applying the whole Endocrine pharmacology toolkit.** The shared principle is negative feedback suppression by exogenous hormone. For each patient, name a first-line approach AND the single biggest safety concern. S29; S29 (notes); S30 (notes)

How it works

- **1.** The hypothalamic-pituitary-target gland axis is a shared template across thyroid, adrenal, and gonadal pharmacology — exogenous hormone therapy suppresses the levels above it via negative feedback. S31
- **2.** Antithyroid drug selection (methimazole vs. PTU) and glucocorticoid tapering both hinge on understanding exactly where and how each drug intervenes in its pathway. S31
- **3.** Glucose-lowering drugs act at distinct sites (pancreas, liver, muscle/fat, kidney, gut, incretin axis) — drug selection balances efficacy against hypoglycemia and other class-specific risks. S31
- **4.** Hyperparathyroidism and hypoparathyroidism require opposite drug strategies (lowering vs. raising effective PTH activity) — a new angle distinct from, but related to, the osteoporosis pharmacotherapy of Unit 2. S31
- **5.** Endocrine pharmacology is deeply interconnected — spironolactone, dopamine agonists, and SERMs each reappear across more than one organ system in this unit alone. S31
- Connecting back: osteoporosis pharmacology (Unit 2) and drug-metabolism/interaction concepts (Unit 1) both reappear here — pharmacology is cumulative, not siloed by unit. S31 (notes)

Drugs named as examples

- **A cumulative self-test spanning both sessions — use this to review before exams.** The recall quiz is a standalone resource, separate from in-lecture quick checks. S32; S32 (notes); S35 (notes)
- **Recall Q1:** PTU blocks both new thyroid hormone synthesis and peripheral T4-to-T3 conversion, giving it a faster effect on circulating active hormone than methimazole — important in thyroid storm. S33; S35
- **Recall Q2:** a patient on chronic glucocorticoid therapy abruptly stops the medication and develops hypotension and weakness: HPA-axis suppression leaves endogenous cortisol production unable to resume quickly. S33; S35
- Chronic exogenous glucocorticoid exposure suppresses CRH/ACTH via negative feedback (Figure 9), reducing the adrenal cortex's stimulus; abrupt withdrawal does not give the axis time to recover, precipitating adrenal insufficiency. S35
- **Recall Q3:** smoking at 35 or older contraindicates estrogen-containing contraception (thromboembolic and cardiovascular risk). **Q4:** sulfonylureas force K-ATP closure regardless of glucose; incretins act only when glucose is high. S33; S34; S35
- **Recall Q4 distractor:** DPP-4 inhibitors do affect insulin secretion — they prolong GLP-1's glucose-dependent effect. S34; S35
- **Recall Q5:** a patient with hypoparathyroidism after thyroid surgery needs treatment — the mainstay is calcitriol plus oral calcium: because PTH is required to activate vitamin D, calcitriol (already-active vitamin D) bypasses that activation step. S34; S35; SG U4-R13
- **Recall Q6:** SGLT2 inhibitors block renal glucose reabsorption, increasing urinary glucose excretion and creating a favorable environment for genital fungal overgrowth

— a mechanism-based adverse effect. S34; S35; SG U4-R12

CLINICAL

- **Patient A — Graves disease, in thyroid storm:** PTU (blocks synthesis and peripheral T4-to-T3 conversion); check liver toxicity. S30; S35
- **Patient B — chronic prednisone user, abruptly out of refills:** restart the glucocorticoid and taper; the risk is adrenal insufficiency from HPA suppression. S30; S35
- **Patient C — type 2 diabetes, older adult with fall history, needs add-on therapy:** DPP-4 or SGLT2 inhibitor; avoid sulfonylurea hypoglycemia. S30; S21
- **Patient D — post-thyroidectomy, perioral numbness and low calcium:** calcitriol plus calcium; monitor for hypercalcemia. S30; S27

TRAP

- Choosing the most potent glucose-lowering drug instead of the safest for an older adult who falls. S21 (notes)
- Integrated case — diabetes: a DPP-4 or SGLT2 inhibitor is preferable to a sulfonylurea given the fall history — sulfonylureas force glucose-independent insulin release, raising hypoglycemia-driven fall risk. SG U4-S
- Integrated case — thyroid storm: beta-blocker + PTU first, then iodine solution only after PTU (never before — iodine first would supply substrate for new hormone synthesis before synthesis is blocked, worsening the crisis). SG U4-S
- Corticosteroids also help in thyroid storm by blunting peripheral conversion and covering relative adrenal insufficiency. SG U4-S
- Shared concept: effective endocrine drug selection depends on knowing exactly where in a pathway (or which receptor/site) a drug acts — that position determines both its efficacy and its specific risks. SG U4-S

Which drug would you choose? (S30)

Patient	Approach	Biggest safety check
A · Graves disease, thyroid storm	PTU	Liver toxicity
B · Chronic prednisone, out of refills	Restart and taper the glucocorticoid	Adrenal insufficiency
C · Older adult with type 2 diabetes and falls	DPP-4 or SGLT2 inhibitor	Avoid sulfonylurea hypoglycemia
D · Post-thyroidectomy, low calcium	Calcitriol + calcium	Hypercalcemia

How to tell them apart. Sources: S30; S35

Supporting drugs at a glance (S67)

Drug(s)	Why it matters here
Rifampin, carbamazepine, phenytoin	CYP inducers that lower contraceptive efficacy
Paroxetine and fluoxetine	CYP2D6 inhibitors that blunt tamoxifen activation
Glucagon and dextrose	Rescue for severe hypoglycemia
Beta-blockers	Can mask hypoglycemia symptoms
Iodinated contrast	Hold metformin in reduced kidney function
Gemfibrozil	Raises repaglinide levels (CYP2C8)
Bisphosphonates and denosumab	Osteoporosis agents covered in Unit 2
Thiazide and loop diuretics	Thiazides raise calcium, loops lower it
Vitamin D (cholecalciferol)	Replaces or supports calcitriol in deficiency

How to tell them apart. Sources: S67; S67 (notes)

MNEMONIC (MEMORY AID)

- Every case, two answers: **the approach** and **the one thing to check**. S30 (notes)

HIGH-YIELD

- Five takeaways: shared axis template · know where each drug intervenes · glucose drugs by site · opposite parathyroid strategies · cross-system drugs. S31

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Estrogen contraindications	VTE/stroke, smoker ≥ 35, migraine with aura, estrogen-sensitive cancer, uncontrolled HTN, liver disease S5
Contraception + rifampin	Backup contraception S6; S67
Tamoxifen	Breast antagonist; endometrial agonist; CYP2D6 inhibitors blunt it S7; S8
Raloxifene	Bone agonist; endometrium blocked; VTE; not for breast cancer treatment S7; S8
Finasteride	↓ DHT; PSA halved → double it; teratogenic tablets S9
Testosterone	Monitor hematocrit, PSA, lipids S7; S9
Insulin buckets	Rapid lispro/aspart · short regular · NPH · basal glargine/detemir/degludec S12
Only IV insulin	Regular S12 (notes)
Hypoglycemia	Rule of 15; beta-blockers mask it; glucagon/dextrose rescue S13; S67
Metformin	AMPK, ↓ gluconeogenesis; eGFR < 30 avoid; contrast hold; B12 S15
Sulfonylurea risk	Highest oral hypoglycemia (forced K-ATP closure) S14; S35
TZD risk	Fluid retention, heart failure S14
SGLT2 risks	Genital yeast, volume depletion, euglycemic DKA, Fournier S16
GLP-1 RA	Weight loss; pancreatitis; no MTC/MEN2; never with DPP-4 S18; S19
Cinacalcet	CaSR activator; hypocalcemia, QT; CYP2D6 inhibitor S26
Hypoparathyroidism	Calcitriol + calcium; watch hypercalcemia S25; S35
Diabetes diagnosis	FPG ≥ 126 or HbA1c ≥ 6.5%, repeated p. 790
Glargine	pH 4.0 — never mix p. 793
Metformin by eGFR	45–60 full · 30–45 caution · < 30 contraindicated p. 798
TZD contraindication	NYHA III–IV heart failure p. 799
Acarbose hypoglycemia	Treat with glucose, not sucrose p. 799

Self-test

First, the lecture's own 6 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S33; S35

Why is PTU, rather than methimazole, preferred in the management of thyroid storm?

- **A** PTU is better absorbed orally
- **B** PTU additionally blocks peripheral T4-to-T3 conversion
- **C** PTU has no hepatotoxicity risk
- **D** PTU works faster at the level of iodide trapping

ANSWER: B

PTU blocks both new thyroid hormone synthesis and peripheral T4-to-T3 conversion, giving it a faster effect on circulating active hormone than methimazole — important in the acute setting of thyroid storm. S33; S35

Lecture check 2 · S33; S35

A patient on chronic glucocorticoid therapy abruptly stops the medication and develops hypotension and weakness. What is the underlying mechanism?

- **A** Direct adrenal gland destruction by the drug
- **B** HPA-axis suppression leaving endogenous cortisol production unable to resume quickly
- **C** An allergic reaction to the glucocorticoid
- **D** Excess aldosterone production

ANSWER: B

Chronic exogenous glucocorticoid exposure suppresses CRH/ACTH via negative feedback (Figure 9), reducing the adrenal cortex’s stimulus to produce cortisol; abrupt withdrawal does not give the axis time to recover, precipitating adrenal insufficiency. S33; S35

Lecture check 3 · S33; S35

Which contraceptive counseling point reflects a true contraindication to estrogen-containing combined hormonal contraception?

- **A** Age under 20
- **B** Smoking at age 35 or older
- **C** Mild seasonal allergies
- **D** A family history of type 2 diabetes

ANSWER: B

Smoking at age 35 or older is a well-established contraindication to estrogen-containing contraception due to substantially increased thromboembolic and cardiovascular risk. S33; S35

Lecture check 4 · S34; S35

Why do sulfonylureas carry a higher hypoglycemia risk than DPP-4 inhibitors?

- **A** Sulfonylureas force K-ATP channel closure and insulin release independent of glucose level, unlike the glucose-dependent incretin pathway
- **B** DPP-4 inhibitors have no effect on insulin secretion
- **C** Sulfonylureas work only in type 1 diabetes
- **D** There is no difference in hypoglycemia risk between these classes

ANSWER: A

Sulfonylureas bind the K-ATP channel directly and force closure regardless of glucose level (Figure 15), while incretin-based agents amplify insulin release only when glucose is already elevated — explaining the difference in hypoglycemia risk. S34; S35

Lecture check 5 · S34; S35

A patient with hypoparathyroidism after thyroid surgery needs pharmacologic treatment. Which combination is the mainstay of therapy?

- **A** Cinacalcet alone
- **B** Calcitriol plus oral calcium
- **C** Bisphosphonate plus denosumab
- **D** Levothyroxine plus methimazole

ANSWER: B

Because PTH is required to activate vitamin D, hypoparathyroidism impairs that activation step; calcitriol (already-active vitamin D) bypasses it, and is given together with oral calcium as first-line therapy. S34; S35

Lecture check 6 · S34; S35

A patient taking an SGLT2 inhibitor develops a genital yeast infection. What is the mechanistic explanation?

- **A** Direct immunosuppression
- **B** Increased urinary glucose excretion promoting fungal overgrowth
- **C** Decreased renal blood flow
- **D** Systemic candidemia from the drug itself

ANSWER: B

SGLT2 inhibitors block renal glucose reabsorption, increasing urinary glucose concentration and creating a favorable environment for genital fungal overgrowth — a mechanism-based, well-known adverse effect. S34; S35

Original questions

Q1 · Clinical vignette · Gonadal Hormones & Inhibitors

A woman on a combined pill starts rifampin for latent TB. What counseling is needed?

- **A** Use backup contraception
- **B** No change; rifampin has no effect
- **C** Double the pill each day
- **D** Stop the pill during rifampin
- **E** Switch rifampin to St. John's wort

ANSWER: A

Enzyme inducers (rifampin, some antiepileptics, St. John's wort) lower contraceptive efficacy (S6). S6; S67

- **B.** It lowers efficacy (S6).
- **C.** Not the advice (S6).
- **D.** Backup, not stopping (S6).
- **E.** Also an inducer (S6).

Q2 · Mechanism · Gonadal Hormones & Inhibitors

Which statement about tamoxifen and raloxifene is correct?

- **A** Tamoxifen weakens bone
- **B** Raloxifene treats breast cancer
- **C** Both are breast agonists
- **D** Only raloxifene raises VTE risk
- **E** Only tamoxifen activates the endometrium

ANSWER: E

Tamoxifen is an endometrial agonist (cancer risk); raloxifene blocks the endometrium (S7–S8). S7; S8

- **A.** Bone agonist (S8).
- **B.** Not used to treat it (S8).
- **C.** Both block breast (S8).
- **D.** Both do (S8).

Q3 · Clinical vignette · Gonadal Hormones & Inhibitors

A man on finasteride for BPH has a PSA of 2.0 ng/mL. How should it be interpreted?

- **A** PSA is meaningless on finasteride
- **B** Stop finasteride for a month first
- **C** Double it — about 4.0 — when screening
- **D** Take it at face value
- **E** Halve it to about 1.0

ANSWER: C

Finasteride lowers PSA by about half, so double the value (S9). S9

- **A.** Still usable, doubled (S9).
- **B.** Not the taught approach (S9).
- **D.** It is halved (S9).
- **E.** Doubled, not halved (S9).

Q4 · Clinical vignette · Insulin

A patient on insulin and propranolol has a glucose of 45 mg/dL but no tremor or palpitations. Why?

- **A** The meter is reading falsely low
- **B** Hypoglycemia never causes tremor
- **C** Propranolol raises glucose
- **D** Insulin is no longer working
- **E** Beta-blockade masks warning symptoms

ANSWER: E

Beta-blockers can mask hypoglycemia warning symptoms (S13; S67). S13; S67

- **A.** Not the taught answer (S13).
- **B.** Tremor is a warning sign masked here (S13).
- **C.** Not the reason (S13).
- **D.** Glucose is low (S13).

Q5 · Mechanism · Insulin

Which insulin is the only one given intravenously?

- **A** Insulin glargine
- **B** Insulin degludec
- **C** NPH insulin
- **D** Regular insulin
- **E** Insulin detemir

ANSWER: D

Only regular insulin is given IV (S12 notes). S12; S47

- **A.** Basal, SC (S12).
- **B.** Basal, SC (S12).
- **C.** Cloudy suspension, SC (S12).
- **E.** Basal, SC (S12).

Q6 · Clinical vignette · Insulin

A patient on insulin is shaky with glucose 58 mg/dL and can swallow. What first?

- **A** 1 mg glucagon IM before any carbs
- **B** 15 g fast carbs, recheck in 15 min
- **C** Wait one hour, then recheck
- **D** Give the next insulin dose early
- **E** Skip the next meal and rest

ANSWER: B

Rule of 15 (S13). S13

- **A.** For patients who cannot take oral carbs (S67).
- **C.** Treat now (S13).
- **D.** Worsens it (S13).
- **E.** Treat now (S13).

Q7 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient on metformin with eGFR 40 is scheduled for a CT with iodinated contrast. What should happen?

- **A** Switch to glyburide permanently
- **B** Add an SGLT2 inhibitor the same day
- **C** Continue without change
- **D** Hold metformin around the contrast study
- **E** Double metformin to cover the stress

ANSWER: D

Hold for contrast dye with reduced kidney function — lactic acidosis risk (S15; S67). S15; S67

- **A.** Not the taught step (S15).
- **B.** Not the answer (S16).
- **C.** Hold (S15).
- **E.** Raises risk (S15).

Q8 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient with a family history of medullary thyroid carcinoma asks about semaglutide for diabetes and weight. Best answer?

- **A** Avoid GLP-1 RAs (C-cell warning)
- **B** Use it with a DPP-4 inhibitor
- **C** Safe if taken with food
- **D** Safe; it only affects the gut
- **E** Only liraglutide is a concern

ANSWER: A

GLP-1 RAs are contraindicated with personal or family history of MTC or MEN2 (S19). S18; S19

- **B.** Never combine (S19).
- **C.** Not relevant (S19).
- **D.** C-cell warning (S19).
- **E.** Class warning (S18).

Q9 · Mechanism · Non-Insulin Glucose-Lowering Drugs

Which class most often causes fluid retention and can worsen heart failure?

- **A** GLP-1 agonists (liraglutide)
- **B** DPP-4 inhibitors (sitagliptin)
- **C** SGLT2 inhibitors (empagliflozin)
- **D** Thiazolidinediones (pioglitazone)
- **E** Biguanides (metformin)

ANSWER: D

TZDs: fluid retention/heart failure, weight gain (S14). S14; S56

- **A.** Weight loss (S18).
- **B.** Weight neutral (S18).
- **C.** Benefit heart failure (S16).
- **E.** No fluid retention (S15).

Q10 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient with type 2 diabetes has NYHA class III heart failure. Which drug is contraindicated?

- **A** Metformin
- **B** Semaglutide
- **C** Pioglitazone
- **D** Empagliflozin
- **E** Sitagliptin

ANSWER: C

Thiazolidinediones cause fluid retention and are contraindicated in NYHA class III and IV heart failure (p. 799). p. 799

- **A.** Not the HF contraindication (p. 798).
- **B.** Not contraindicated here (p. 800).
- **D.** Reduces heart-failure events (p. 802).
- **E.** Not contraindicated here (p. 801).

Q11 · Mechanism · Parathyroid Disorders

What should be monitored after starting cinacalcet?

- **A** Glucose for hypoglycemia
- **B** Potassium for hyperkalemia
- **C** TSH for hypothyroidism
- **D** Calcium for hypercalcemia
- **E** Serum calcium for hypocalcemia

ANSWER: E

Cinacalcet lowers PTH and calcium; monitor for hypocalcemia (and QT) (S25–S26). S25; S26

- **A.** Not the taught target (S26).
- **B.** Not the taught target (S26).
- **C.** Not the taught target (S26).
- **D.** That is calcitriol (S25).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO4.1** Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.
- [ ] **IO4.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.
- [ ] **IO4.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO4.4** Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.
- [ ] **IO4.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.
- [ ] **IO4.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.
- [ ] **SO1** Describe the mechanisms and clinical uses of gonadal hormone agents, including combined hormonal contraception, SERMs, and androgens/antiandrogens.

- [ ] **SO2** Classify glucose-lowering drugs by mechanism and site of action, and compare agents on efficacy, hypoglycemia risk, and key adverse effects.
- [ ] **SO3** Distinguish hyperparathyroidism from hypoparathyroidism and identify the drug strategy used for each.
- [ ] **SO4** Identify key contraindications, adverse effects, and drug interactions throughout.
- [ ] **SO5** Apply the full Endocrine pharmacology toolkit to integrated clinical scenarios, and complete a cumulative recall quiz spanning both sessions.